

VARIVAX® Refrigerated
Varicella Virus Vaccine, Live (Oka/Merck)

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LOCAL PRODUCT CIRCULAR

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Varicella Virus Vaccine, Live (Oka/Merck)

I. THERAPEUTIC CLASS

VARIVAX® Refrigerated is a live, attenuated virus vaccine (a lyophilized preparation of the Oka/Merck strain of varicella).

II. CHEMISTRY

VARIVAX Refrigerated is a lyophilized preparation of the Oka/Merck strain of live, attenuated varicella virus. The virus was initially obtained from a child with wild-type varicella, then introduced into human embryonic lung cell cultures, adapted to and propagated in embryonic guinea pig cell cultures and finally propagated in human diploid cell cultures (WI-38). Further passage of the virus for varicella vaccine was performed at Research Laboratories of Merck Sharp & Dohme, LLC, Rahway, NJ, USA in human diploid cell cultures (MRC-5) that were free of adventitious agents. This live, attenuated varicella vaccine is a lyophilized preparation containing sucrose, phosphate, glutamate, processed gelatin, and urea as stabilizers.

III. COMPOSITION

IIIa. Active Ingredients

VARIVAX Refrigerated, when reconstituted as directed, is a sterile preparation for intramuscular or subcutaneous administration. Each approximately 0.5 mL dose contains: a minimum of 1350 PFU (plaque forming units) of Oka/Merck varicella virus when reconstituted and stored at room temperature for 30 minutes.

IIIb. Inactive Ingredients

Each approximately 0.5 mL dose contains: approximately 17 mg of sucrose, 8.3 mg of hydrolyzed gelatin, 3.5 mg of urea, 2.1 mg of sodium chloride, 0.33 mg of monosodium L-glutamate, 0.30 mg of sodium phosphate dibasic, 53 mcg of potassium phosphate monobasic, 53 mcg of potassium chloride. The product also contains residual components of MRC-5 cells and trace quantities of neomycin and bovine calf serum from MRC-5 culture media. The product contains no preservative.

IV. CLINICAL PHARMACOLOGY

IVa. Varicella

Varicella is a highly communicable disease in children, adolescents, and adults caused by the varicella-zoster virus. The disease usually consists of 300 to 500 maculopapular and/or vesicular lesions accompanied by a fever (oral temperature $\geq 100^{\circ}\text{F}$) in up to 70% of individuals. Approximately 3.5 million cases of varicella occurred annually from 1980 to 1994 in the United States with the peak incidence occurring in children five to nine years of age. The incidence rate of varicella was 8.3 to 9.1% per year in children 1 to 9 years of age. The attack rate of wild-type varicella following household exposure among healthy susceptible children was shown to be 87%. Although it is generally a benign, self-limiting disease, varicella may be associated with serious complications (e.g., bacterial superinfection, pneumonia, encephalitis, Reye syndrome), and/or death.

IVb. Clinical Studies

Evaluation of Clinical Efficacy Afforded by VARIVAX Refrigerated

The protective efficacy of varicella virus vaccine live (Oka/Merck) [hereafter referred to as varicella vaccine (Oka/Merck)] administered subcutaneously was established by: (1) a placebo-controlled, double-blind clinical trial, (2) comparing varicella rates in vaccinees versus historical controls, and (3) assessing protection from disease following household exposure.

Clinical Data in Children

One-Dose Regimen in Children

Although no placebo-controlled trial was carried out with refrigerator-stable varicella vaccine (Oka/Merck), a placebo-controlled trial was conducted using a prior formulation containing 17,000 PFU per dose. In this trial, a single dose of varicella vaccine (Oka/Merck) protected 96 to 100% of children against varicella over a two-year period. The study enrolled healthy individuals 1 to 14 years of age (n=491 vaccine, n=465 placebo). In the first year, 8.5% of placebo recipients contracted varicella, while no vaccine recipient did, for a calculated protection rate of 100% during the first varicella season. In the second year, when only a subset of individuals agreed to remain in the blinded study (n=163 vaccine, n=161 placebo), 96% protective efficacy was calculated for the vaccine group as compared with placebo.

In early clinical trials, a total of 4240 children 1 to 12 years of age received 1000 to 1625 PFU of attenuated virus per dose of varicella vaccine (Oka/Merck) and have been followed for up to nine years post single-dose vaccination. In this group, there was considerable variation in varicella rates among studies and study sites, and much of the reported data was acquired by passive follow-up. It was observed that 0.3 to 3.8% of vaccinees per year reported varicella (called breakthrough cases). This represents an approximate 83% (95% confidence interval [CI], 82%, 84%) decrease from the age-adjusted expected incidence rates in susceptible subjects over this same period. In those who developed breakthrough varicella postvaccination, the majority experienced mild disease (median of the maximum number of lesions <50). In one study, a total of 47% (27/58) of breakthrough cases had <50 lesions compared with 8% (7/92) in unvaccinated individuals, and 7% (4/58) of breakthrough cases had >300 lesions compared with 50% (46/92) in unvaccinated individuals.

Among a subset of vaccinees who were actively followed in these early trials for up to nine years postvaccination, 179 individuals had household exposure to varicella. There were no reports of breakthrough varicella in 84% (150/179) of exposed children, while 16% (29/179) reported a mild form of varicella (38% [11/29] of the cases with a maximum total number of <50 lesions; no individuals with >300 lesions). This represents an 81% reduction in the expected number of varicella cases utilizing the historical attack rate of 87% following household exposure to varicella in unvaccinated individuals in the calculation of efficacy.

In later clinical trials, a total of 1114 children 1 to 12 years of age received 2900 to 9000 PFU of attenuated virus per dose of varicella vaccine (Oka/Merck) and have been actively followed for up to 10 years post single-dose vaccination. It was observed that 0.2 to 2.3% of vaccinees per year reported breakthrough varicella for up to 10 years post single-dose vaccination. This represents an estimated efficacy 94% (95% CI, 93%, 96%) compared with the age-adjusted expected incidence rates in susceptible subjects over the same period. In those who developed breakthrough varicella postvaccination, the majority experienced mild disease with the median of the maximum total number of lesions <50. The severity of reported breakthrough varicella, as measured by number of lesions and maximum temperature, appeared not to increase with time since vaccination.

Among a subset of vaccinees who were actively followed in these later trials for up to 10 years postvaccination, 95 individuals were exposed to an unvaccinated individual with wild-type varicella in a household setting. There were no reports of breakthrough varicella in 92% (87/95) of exposed children, while 8% (8/95) reported a mild form of varicella (maximum total number of lesions <50; observed range, 10 to 34). This represents an estimated efficacy of 90% (95% CI, 82%, 96%) based on the historical attack rate of 87% following household exposure to varicella in unvaccinated individuals in the calculation of efficacy.

Two-Dose Regimen in Children

In a clinical trial, a total of 2216 children 12 months to 12 years of age with a negative history of varicella were randomized to receive either 1 dose of varicella vaccine (Oka/Merck) (n=1114) or 2 doses of varicella vaccine (Oka/Merck) (n=1102) given 3 months apart. Subjects were actively followed for varicella, any varicella-like illness, or herpes zoster and any exposures to varicella or herpes zoster on an annual basis for 10 years after vaccination. Persistence of varicella-zoster

vaccine (VZV) antibody was measured annually for 9 years. Most cases of varicella reported in recipients of 1 dose or 2 doses of vaccine were mild. The estimated vaccine efficacy for the 10-year observation period was 94% for 1 dose and 98% for 2 doses ($p < 0.001$). This translates to a 3.4-fold lower risk of developing varicella >42 days postvaccination during the 10-year observation period in children who received 2 doses than in those who received 1 dose (2.2% vs. 7.5%, respectively).

Clinical Data in Adolescents and Adults

Although no placebo-controlled trial was carried out in adolescents and adults, the protective efficacy of varicella vaccine (Oka/Merck) was calculated by evaluation of protection when vaccinees received 2 doses of varicella vaccine (Oka/Merck) 4 or 8 weeks apart and were subsequently exposed to varicella in a household setting over 6 to 7 years. In earlier clinical trials with up to 6 years of follow-up, 13 of the 76 individuals (17%) who had household exposure to varicella, developed varicella. All of the varicella cases that were reported were generally mild with a median of 37 lesions (range 8 to 75). In later clinical trials with up to 7 years of follow-up, none of 19 individuals (0%) who had household exposure developed varicella.

The attack rate of unvaccinated adults exposed to a single contact in a household has not been previously studied. If the attack rate of 87% following household exposure in susceptible children holds true for susceptible adolescents and adults, the estimated efficacy of the vaccine in the prevention of any varicella disease would range from 80 to 100%.

There are an insufficient number of breakthrough varicella cases among vaccinated adolescents and adults to assess the rate of protection of varicella vaccine (Oka/Merck) against the serious complications of varicella (e.g., encephalitis, hepatitis, pneumonia) and during pregnancy (congenital varicella syndrome).

Immunogenicity of Varicella Vaccine (Oka/Merck)

In clinical trials, varicella antibodies have been evaluated following vaccination with formulations of varicella vaccine (Oka/Merck) containing attenuated virus ranging from 1000 to 50,000 PFU per dose in healthy individuals ranging from 12 months to 55 years of age.

One-Dose Regimen in Children

In prelicensure efficacy studies, seroconversion was observed in 97% of vaccinees at approximately 4 to 6 weeks postvaccination in 6889 susceptible children 12 months to 12 years of age. Titers ≥ 5 gpELISA units/mL were induced in approximately 76% of children vaccinated with a single dose of vaccine at 1000 to 17,000 PFU per dose. Rates of breakthrough disease were significantly lower among children with VZV antibody titers ≥ 5 gpELISA units/mL compared with children with titers < 5 gpELISA units/mL.

Immunogenicity of refrigerator-stable varicella vaccine (Oka/Merck) (6550 PFU per dose, n=320 and 28,400 PFU per dose, n=315) administered subcutaneously was compared with that of the licensed frozen formulation of varicella vaccine (Oka/Merck) (9189 PFU per dose, n=323) administered subcutaneously in a double-blind, randomized, multicenter study in U.S. children 12 to 23 months of age, all of whom received M-M-R II (Measles, Mumps, and Rubella Virus Vaccine Live) concomitantly. The per-protocol analysis included all subjects with prevaccination varicella antibody titers < 1.25 gpELISA units (n=267 to 276 per group); the antibody responses were comparable across the 3 treatment groups, with 6-week postvaccination varicella antibody titers ≥ 5 gpELISA units in 93.3%, 93.8%, and 95.1% of subjects, respectively.

In an open label clinical trial (NCT00432523) 752 children 12 through 18 months of age received VARIVAX either intramuscularly (n=374) or subcutaneously (n=378), concomitantly with M-M-R II. Antibody responses to varicella virus were measured by gpELISA using sera obtained 6 weeks postvaccination. Seroresponse rate was defined as the percentage of initially seronegative children who achieved antibody titers above the seroresponse threshold for the assay 6 weeks postvaccination. The seroresponse threshold was defined as 5 gpELISA units for anti-varicella virus antibodies. Ninety-five percent of enrolled children were seronegative to varicella virus at baseline. In the pre-specified primary analysis, seroresponse rates to varicella virus were noninferior in the intramuscular group compared to the subcutaneous group (the lower bound of the 95% confidence interval for the difference in seroresponse rates [intramuscular group minus subcutaneous group] was $\geq -10\%$). The proportions of children achieving antibody titers above the seroresponse thresholds for varicella virus were 88.4% (95% CI: 84.5, 91.6) of children in the intramuscular group and 85.5% (95% CI: 81.3, 89.0) of children in the subcutaneous.

Two-Dose Regimen in Children

In a multicenter study, 2216 healthy children 12 months to 12 years of age received either 1 dose or 2 doses of varicella vaccine (Oka/Merck) administered subcutaneously 3 months apart. The immunogenicity results are shown in the following table.

Table 1: Summary of VZV Antibody Responses at 6 Weeks Postdose 1 and 6 Weeks Postdose 2 in Initially Seronegative Children 12 Months to 12 Years of Age (Vaccinations 3 Months Apart)

	Varicella Vaccine (Oka/Merck) 1-Dose Regimen (N=1114)	Varicella Vaccine (Oka/Merck) 2-Dose Regimen (3 months apart) (N=1102)	
	6 Weeks Postvaccination (n=892)	6 Weeks Postdose 1 (n=851)	6 Weeks Postdose 2 (n=769)
Seroconversion Rate	98.9%	99.5%	99.9%
Percent with VZV Antibody Titer ≥ 5 gpELISA units/mL	84.9%	87.3%	99.5%
Geometric mean titers (gpELISA units/mL) (95% CI)	12.0 (11.2, 12.8)	12.8 (11.9, 13.7)	141.5 (132.3, 151.3)

N = Number of subjects vaccinated.

n = Number of subjects included in immunogenicity analysis.

The results from this study and other studies in which a second dose of varicella vaccine (Oka/Merck) was administered 3 to 6 years after the initial dose demonstrate significant boosting of the VZV antibodies with a second dose. VZV antibody levels after 2 doses given 3 to 6 years apart are comparable to those obtained when the 2 doses are given 3 months apart.

Two-Dose Regimen in Adolescents and Adults

In a multicenter study involving susceptible adolescents and adults 13 years of age and older, two doses of varicella vaccine (Oka/Merck) administered subcutaneously four to eight weeks apart induced a seroconversion rate (gpELISA ≥ 0.6 units) of approximately 75% in 539 individuals four weeks after the first dose and of 99% in 479 individuals four weeks after the second dose.

The average antibody response in vaccinees who received the second dose eight weeks after the first dose was higher than that in those who received the second dose four weeks after the first dose. In another multicenter study involving adolescents and adults, two doses of varicella vaccine (Oka/Merck) administered subcutaneously eight weeks apart induced a seroconversion rate (gpELISA ≥ 0.6 units) of 94% in 142 individuals six weeks after the first dose and 99% in 122 individuals six weeks after the second dose.

Varicella vaccine (Oka/Merck) also induces cell-mediated immune responses in vaccinees. The relative contributions of humoral immunity and cell-mediated immunity to protection from varicella are unknown.

Persistence of Immune Response

One-Dose Regimen in Children

In clinical studies involving healthy children who received 1 dose of VARIVAX subcutaneously, detectable VZV antibodies were present in 99.0% (3886/3926) at 1 year, 99.3% (1555/1566) at 2 years, 98.6% (1106/1122) at 3 years, and 99.4% (1168/1175) at 4 years, 99.2% (737/743) at 5 years, 100% (142/142) at 6 years, 97.4% (38/39) at 7 years, 100% (34/34) at 8 years, and 100% (16/16) at 10 years postvaccination.

Two-Dose Regimen in Children

In recipients of 1 dose of VARIVAX administered subcutaneously over 9 years of follow-up, the geometric mean titers (GMTs) and the percent of subjects with VZV antibody titers ≥ 5 gpELISA units/mL generally increased. The GMTs and percent of subjects with VZV antibody titers ≥ 5 gpELISA units/mL in the 2-dose recipients were higher than those in the 1-dose recipients for the first year follow-up and generally comparable thereafter. The cumulative rate of VZV antibody persistence with both regimens remained very high at year 9 (99.0% for the 1-dose group and 98.8% for the 2-dose group).

Two-Dose Regimen in Adolescents and Adults

In clinical studies involving healthy adolescents and adults who received 2 doses of VARIVAX subcutaneously, detectable varicella antibodies (gpELISA ≥ 0.6 units) were present in 97.9% (568/580) at 1 year, 97.1% (34/35) at 2 years, 100% (144/144) at 3 years, 97.0% (98/101) at 4 years, 97.5% (78/80) at 5 years, and 100% (45/45) at 6 years postvaccination.

A boost in antibody levels has been observed in vaccinees following exposure to wild-type varicella which could account for the apparent long-term persistence of antibody levels after vaccination in these studies. The duration of protection from varicella obtained using varicella vaccine (Oka/Merck) in the absence of wild-type boosting is unknown. Varicella vaccine (Oka/Merck) also induces cell-mediated immune responses in vaccinees. The relative contributions of humoral immunity and cell-mediated immunity to protection from varicella are unknown.

Vaccination with VARIVAX (Refrigerated) may not result in protection of all healthy, susceptible children, adolescents, and adults.

Transmission

In the placebo-controlled trial, transmission of vaccine virus was assessed in household settings (during the 8-week postvaccination period) in 416 susceptible placebo recipients who were household contacts of 445 vaccine recipients. Of the 416 placebo recipients, three developed varicella and seroconverted, nine reported a varicella-like rash and did not seroconvert, and six had no rash but seroconverted. If vaccine virus transmission occurred, it did so at a very low rate and possibly without recognizable clinical disease in contacts. These cases may represent either wild-type varicella from community contacts or a low incidence of transmission of vaccine virus from vaccinated contacts. Post-marketing experience suggests that transmission of varicella vaccine virus (Oka/Merck) resulting in varicella infection including disseminated disease may occur rarely between vaccine recipients (who develop or do not develop a varicella-like rash) and contacts susceptible to varicella including healthy as well as high-risk individuals (see **PRECAUTIONS, Transmission**).

Herpes Zoster

Overall, 9543 healthy children (12 months to 12 years of age) and 1652 adolescents and adults (13 years of age and older) have been vaccinated with varicella vaccine (Oka/Merck) in clinical trials. Twelve cases of herpes zoster have been reported in children during 84,414 person years of follow-up in clinical trials, resulting in a calculated incidence of at least 14 cases per 100,000 person years. The completeness of this reporting has not been determined. Two cases of herpes

zoster have been reported in the adolescent and adult age group during 12,372 person years of follow-up in clinical trials resulting in a calculated incidence of 16 cases per 100,000 person years.

All 14 cases of herpes zoster were mild and no sequelae were reported. Two cultures (one child and one adult) obtained from vesicles were positive for wild-type varicella-zoster virus as confirmed by restriction endonuclease analysis. The long-term effect of varicella vaccine (Oka/Merck) on the incidence of herpes zoster, particularly in those vaccinees exposed to wild-type varicella, is unknown at present.

In children, the reported rate of zoster in vaccine recipients appears not to exceed that previously determined in a population-based study of healthy children who had experienced wild-type varicella. The incidence of zoster in adults who have had wild-type varicella infection is higher than that in children.

Reye Syndrome

Reye syndrome has occurred in children and adolescents following wild-type varicella infection, the majority of whom had received salicylates. In clinical studies in healthy children and adolescents in the United States, physicians advised varicella vaccine recipients not to use salicylates for six weeks after vaccination. There were no reports of Reye syndrome in varicella vaccine recipients during these studies.

Studies with Other Vaccines

In combined clinical studies involving 1107 children 12 to 36 months of age, 680 children received varicella vaccine (Oka/Merck) and M-M-R II* (Measles, Mumps, and Rubella Virus Vaccine Live) administered subcutaneously and concomitantly at separate sites and 427 received the vaccines six weeks apart. Seroconversion rates and antibody levels were comparable between the two groups at approximately six weeks postvaccination to each of the virus vaccine components. No differences were noted in adverse reactions reported in those who received varicella vaccine (Oka/Merck) concomitantly with M-M-R II at separate sites and those who received varicella vaccine (Oka/Merck) and M-M-R II at different times (see **DRUG INTERACTIONS**).

In a clinical study involving 316 children 12 months to 42 months of age, 160 received an investigational vaccine (a formulation combining measles, mumps, rubella, and varicella in one

syringe) administered subcutaneously and concomitantly with booster doses of DTaP (diphtheria, tetanus, acellular pertussis) and OPV (oral poliovirus vaccine) while 156 received M-M-R II concomitantly with booster doses of DTaP and OPV followed by varicella vaccine (Oka/Merck) administered subcutaneously 6 weeks later. At six weeks postvaccination, seroconversion rates for measles, mumps, rubella, and varicella and the percentage of vaccinees whose titers were boosted for diphtheria, tetanus, pertussis, and polio were comparable between the two groups, but anti-varicella levels were decreased when the investigational vaccine containing varicella was administered concomitantly with DTaP. No clinically significant differences were noted in adverse reactions between the two groups.

In another clinical study involving 306 children 12 to 18 months of age, 151 received an investigational vaccine (a formulation combining measles, mumps, rubella, and varicella in one syringe) administered subcutaneously and concomitantly with a booster dose of PedvaxHIB [*Haemophilus b* Conjugate Vaccine (Meningococcal Protein Conjugate)] while 155 received M-M-R II concomitantly with a booster dose of PedvaxHIB followed by varicella vaccine (Oka/Merck) administered subcutaneously 6 weeks later. At six weeks postvaccination, seroconversion rates for measles, mumps, rubella, and varicella, and geometric mean titers for PedvaxHIB were comparable between the two groups, but anti-varicella levels were decreased when the investigational vaccine containing varicella was administered concomitantly with PedvaxHIB. No clinically significant differences in adverse reactions were seen between the two groups.

In a clinical study involving 609 children 12 months to 23 months of age, 305 received varicella vaccine (Oka/Merck), M-M-R II, and *Haemophilus influenzae* type b, diphtheria, tetanus, and pertussis combined vaccine concomitantly at separate sites and 304 received M-M-R II and *Haemophilus influenzae* type b, diphtheria, tetanus, and pertussis combined vaccine given concomitantly followed by varicella vaccine (Oka/Merck) 6 weeks later. At six weeks postvaccination, seroconversion rates for measles, mumps, rubella, and varicella were similar between the two groups. Compared to prevaccination GMTs, the six-week postvaccination boost in GMTs for *Haemophilus influenzae* type b, diphtheria, tetanus and pertussis was similar between the two groups. GMTs for all antigens were similar except for varicella which was lower when varicella vaccine (Oka/Merck) was administered concomitantly with M-M-R II and *Haemophilus influenzae* type b, diphtheria, tetanus, and pertussis combined vaccine but within

the range of GMTs seen in previous clinical experience when varicella vaccine (Oka/Merck) was administered alone. At 1 year postvaccination, GMTs for measles, mumps, rubella, varicella and *Haemophilus influenzae* type b were similar between the two groups. All three vaccines were well tolerated regardless of whether they were administered concomitantly at separate sites or 6 weeks apart. There were no clinically important differences in reaction rates when the three vaccines were administered concomitantly versus 6 weeks apart.

In a clinical study involving 822 children 12 to 15 months of age, 410 received *Haemophilus influenzae* type b Conjugate (Meningococcal Protein Conjugate) and Hepatitis B (Recombinant) combined vaccine, M-M-R II, and varicella vaccine (Oka/Merck) concomitantly at separate sites, and 412 received *Haemophilus influenzae* type b Conjugate (Meningococcal Protein Conjugate) and Hepatitis B (Recombinant) combined vaccine followed by M-M-R II and varicella vaccine (Oka/Merck) given concomitantly at separate sites, 6 weeks later. In this study VARIVAX was administered subcutaneously. At 6 weeks postvaccination, the immune responses for the subjects who received the concomitant injections of *Haemophilus influenzae* type b Conjugate (Meningococcal Protein Conjugate) and Hepatitis B (Recombinant) combined vaccine, M-M-R II, and varicella vaccine (Oka/Merck) were similar to those of the subjects who received *Haemophilus influenzae* type b Conjugate (Meningococcal Protein Conjugate) and Hepatitis B (Recombinant) combined vaccine followed 6 weeks later by M-M-R II and varicella vaccine (Oka/Merck) with respect to all antigens administered. All 3 vaccines were generally well tolerated regardless of whether they were administered concomitantly at separate sites or 6 weeks apart. There were no clinically important differences in reaction rates when the 3 vaccines were administered concomitantly versus 6 weeks apart.

VARIVAX Refrigerated is recommended for intramuscular or subcutaneous administration. During clinical trials, some children received varicella vaccine (Oka/Merck) intramuscularly resulting in seroconversion rates similar to those in children who received the vaccine by the subcutaneous route. Persistence of antibody and efficacy in those receiving intramuscular injections have not been defined.

V. INDICATIONS

VARIVAX Refrigerated is indicated for vaccination against varicella in individuals 12 months of age and older.

Revaccination

The duration of protection of VARIVAX Refrigerated is unknown at present and the need for booster doses is not defined. However, a boost in antibody levels has been observed in vaccinees following exposure to wild-type varicella as well as following a booster dose of varicella virus vaccine live (Oka/Merck) [hereafter referred to as varicella vaccine (Oka/Merck)] administered four to six years postvaccination.

In a highly vaccinated population, immunity for some individuals may wane due to lack of exposure to wild-type varicella as a result of shifting epidemiology. Post-marketing surveillance studies are ongoing to evaluate the need and timing for booster vaccination.

Vaccination with VARIVAX Refrigerated may not result in protection of all healthy, susceptible children, adolescents, and adults (see **CLINICAL PHARMACOLOGY**).

VI. DOSAGE AND ADMINISTRATION

FOR INTRAMUSCULAR OR SUBCUTANEOUS ADMINISTRATION ONLY.

Do not inject intravenously.

Children 12 months to 12 years of age should receive an approximately 0.5 mL dose administered intramuscularly or subcutaneously. If a second dose is administered, there should be a minimum interval of 3 months between doses (see **CLINICAL PHARMACOLOGY**).

Adolescents and adults 13 years of age and older should receive an approximately 0.5 mL dose administered intramuscularly or subcutaneously at elected date and a second approximately 0.5 mL dose 4 to 8 weeks later.

The outer aspect of the upper arm (deltoid region) is the preferred site of injection.

Methods of administration

Prefilled syringe of diluent:

To reconstitute the vaccine, inject all of the diluent (0.7 mL) in the prefilled syringe into the vial of lyophilized vaccine and gently agitate to mix thoroughly. Withdraw the entire contents into the syringe and inject the total volume (approximately 0.5 mL) of reconstituted vaccine intramuscularly or subcutaneously, preferably into the outer aspect of the upper arm (deltoid region) or the anterolateral thigh. **IT IS RECOMMENDED THAT THE VACCINE BE ADMINISTERED IMMEDIATELY AFTER RECONSTITUTION, TO MINIMIZE LOSS OF POTENCY. DISCARD IF RECONSTITUTED VACCINE IS NOT USED WITHIN 30 MINUTES.**

Do not freeze reconstituted vaccine.

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit. VARIVAX Refrigerated when reconstituted is a clear, colorless to pale yellow liquid.

Vial of diluent:

The diluent should be stored separately at room temperature (20 to 25°C, 68 to 77°F), or in the refrigerator.

To reconstitute the vaccine, first withdraw 0.7 mL of diluent into the syringe to be used for reconstitution. Inject all of the diluent in the syringe into the vial of lyophilized vaccine and gently agitate to mix thoroughly. Withdraw the entire contents into a syringe and inject the total volume (approximately 0.5 mL) of reconstituted vaccine intramuscularly or subcutaneously, preferably into the outer aspect of the upper arm (deltoid region) or the anterolateral thigh. **IT IS RECOMMENDED THAT THE VACCINE BE ADMINISTERED IMMEDIATELY AFTER RECONSTITUTION, TO MINIMIZE LOSS OF POTENCY. DISCARD IF RECONSTITUTED VACCINE IS NOT USED WITHIN 30 MINUTES.**

Do not freeze reconstituted vaccine.

CAUTION: A sterile syringe free of preservatives, antiseptics, and detergents should be used for each injection and/or reconstitution of VARIVAX Refrigerated because these substances may inactivate the vaccine virus.

It is important to use a separate sterile syringe and needle for each patient to prevent transmission of infectious agents from one individual to another.

To reconstitute the vaccine, use only the diluent supplied, since it is free of preservatives or other anti-viral substances which might inactivate the vaccine virus.

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit. VARIVAX Refrigerated when reconstituted is a clear, colorless to pale yellow liquid.

VII. CONTRAINDICATIONS

History of hypersensitivity to any component of the vaccine, including gelatin.

History of anaphylactoid reaction to neomycin (each dose of reconstituted vaccine contains trace quantities of neomycin).

Blood dyscrasias, leukemia, lymphomas of any type, or other malignant neoplasms affecting the bone marrow or lymphatic systems.

Immunosuppressive therapy (including high-dose corticosteroids); however, VARIVAX Refrigerated is not contraindicated for use with topical corticosteroids or low-dose corticosteroids, as are commonly used for asthma prophylaxis. Individuals who are on immunosuppressant drugs are more susceptible to infections than healthy individuals. Vaccination with live attenuated varicella vaccine can result in a more extensive vaccine-associated rash or disseminated disease in individuals on immunosuppressant doses of corticosteroids.

Primary and acquired immunodeficiency states, including immunosuppression in association with AIDS or other clinical manifestations of infection with human immunodeficiency virus, except immunosuppression in asymptomatic children with CD4 T-lymphocyte percentages $\geq 25\%$.

Family history of congenital or hereditary immunodeficiency, unless the immune competence of the potential vaccine recipient is demonstrated.

Active untreated tuberculosis.

Any active febrile illness with fever $>38.5^{\circ}\text{C}$ ($>101.3^{\circ}\text{F}$); however, low-grade fever itself is not a contraindication to vaccination.

Pregnancy; the possible effects of the vaccine on fetal development are unknown at this time. However, wild-type varicella is known to sometimes cause fetal harm. If vaccination of postpubertal females is undertaken, pregnancy should be avoided for three months following vaccination (see **PREGNANCY**).

VIII. PRECAUTIONS

Adequate treatment provisions, including epinephrine injection (1:1000), should be available for immediate use should an anaphylactoid reaction occur.

The duration of protection from varicella infection after vaccination with VARIVAX Refrigerated is unknown.

The safety and efficacy of VARIVAX Refrigerated have not been established in children and young adults who are known to be infected with human immunodeficiency virus with and without evidence of immunosuppression (see also **CONTRAINDICATIONS**).

Transmission

Post-marketing experience suggests that transmission of varicella vaccine virus (Oka/Merck) resulting in varicella infection including disseminated disease may occur rarely between vaccine

recipients (who develop or do not develop a varicella-like rash) and contacts susceptible to varicella including healthy as well as high-risk individuals.

Therefore, vaccine recipients should attempt to avoid, whenever possible, close association with susceptible high-risk individuals for up to six weeks. In circumstances where contact with high-risk individuals is unavoidable, the potential risk of transmission of vaccine virus should be weighed against the risk of acquiring and transmitting wild-type varicella virus. Susceptible high-risk individuals include:

- immunocompromised individuals
- pregnant women without documented history of varicella or laboratory evidence of prior infection
- newborn infants of mothers without documented history of varicella or laboratory evidence of prior infection.

IX. PREGNANCY

There are no adequate and well-controlled studies in pregnant women. It is not known whether VARIVAX Refrigerated can cause fetal harm when administered to a pregnant woman or can affect reproduction capacity. Therefore, VARIVAX Refrigerated should not be administered to pregnant females; furthermore, pregnancy should be avoided for three months following vaccination (see **CONTRAINDICATIONS**).

X. NURSING MOTHERS

It is not known whether varicella vaccine virus is secreted in human milk. Therefore, because some viruses are secreted in human milk, caution should be exercised if VARIVAX Refrigerated is administered to a nursing woman.

XI. PEDIATRIC USE

No clinical data are available on safety or efficacy of VARIVAX Refrigerated in children less than one year of age. Administration to infants under twelve months of age is not recommended.

XII. EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

There are no data to suggest that VARIVAX Refrigerated affects the ability to drive or operate machinery.

XIII. DRUG INTERACTIONS

Vaccination should be deferred for at least 5 months following blood or plasma transfusions, or administration of immune globulin or varicella-zoster immune globulin (VZIG).

Following administration of VARIVAX Refrigerated, any immune globulin including VZIG should not be given for 2 months thereafter unless its use outweighs the benefits of vaccination.

Vaccine recipients should avoid use of salicylates for 6 weeks after vaccination with VARIVAX Refrigerated as Reye syndrome has been reported following the use of salicylates during wild-type varicella infection.

Results from clinical studies indicate that VARIVAX Refrigerated can be administered concomitantly with M-M-R[®]II (Measles, Mumps, and Rubella Virus Vaccine Live), diphtheria and tetanus toxoids and pertussis vaccine adsorbed and *Haemophilus influenzae* type b conjugate combined vaccine, or *Haemophilus influenzae* type b Conjugate (Meningococcal Protein Conjugate) and Hepatitis B (Recombinant) combined vaccine. If VARIVAX Refrigerated is not given concomitantly with M-M-R II, a 1-month interval between the 2 live virus vaccines should be observed.

Limited data from an experimental product containing varicella vaccine suggest that VARIVAX Refrigerated can be administered concomitantly with DTaP (diphtheria, tetanus, acellular

pertussis) and PedvaxHIB* [*Haemophilus b* Conjugate Vaccine (Meningococcal Protein Conjugate)] using separate sites and syringes and with OPV (oral poliovirus vaccine).

XIV. SIDE EFFECTS

Clinical Studies

Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a vaccine cannot be directly compared to rates in the clinical trials of another vaccine and may not reflect the rates observed in clinical practice. Vaccine-related adverse reactions reported during clinical trials were assessed by the study investigators to be possibly, probably, or definitely vaccine-related and are summarized below.

In clinical trials, varicella vaccine (Oka/Merck) was administered subcutaneously to over 11,000 healthy children, adolescents, and adults.

In a double-blind, placebo-controlled study among 914 healthy children and adolescents who were serologically confirmed to be susceptible to varicella, the only adverse reactions that occurred at a significantly ($p < 0.05$) greater rate in vaccine recipients than in placebo recipients were pain and redness at the injection site.

Children 1 to 12 Years of Age

One-Dose Regimen in Children

In clinical trials involving healthy children monitored for up to 42 days after a single dose of varicella vaccine (Oka/Merck), the frequency of fever, injection-site complaints, or rashes were reported as shown in Table 2:

Table 2: Fever, Local Reactions, and Rashes (%) in Children 1 to 12 Years of Age 0 to 42 Days After Receipt of a Single Dose of Varicella Vaccine (Oka/Merck)

Reaction	N	% Experiencing Reaction	Peak Occurrence During Postvaccination Days

Fever $\geq 102.0^{\circ}\text{F}$ (38.9°C) Oral	8824	14.7%	0 to 42
Injection-site complaints (pain/soreness, swelling and/or erythema, rash, pruritus, hematoma, induration, stiffness)	8913	19.3%	0 to 2
Varicella-like rash (injection site)	8913	3.4%	8 to 19
Median number of lesions		2	
Varicella-like rash (generalized)	8913	3.8%	5 to 26
Median number of lesions		5	

In addition, adverse events occurring at a rate of $\geq 1\%$ are listed in decreasing order of frequency: upper respiratory illness, cough, irritability, fatigue, disturbed sleep, diarrhea, loss of appetite, vomiting, otitis, headache, malaise, abdominal pain, other rash, nausea, chills, lymphadenopathy, myalgia, lower respiratory illness, allergic reactions (including allergic rash, hives), stiff neck, arthralgia, itching.

Pneumonitis has been reported rarely ($<1\%$) in children vaccinated with varicella vaccine (Oka/Merck).

Febrile seizures have occurred at a rate of $<0.1\%$ in children vaccinated with varicella vaccine (Oka/Merck).

Clinical safety of refrigerator-stable varicella vaccine (Oka/Merck) ($n=635$) was compared with that of the licensed frozen formulation of varicella vaccine (Oka/Merck) ($n=323$) for 42 days postvaccination in U.S. children 12 to 23 months of age. The safety profiles were comparable for the two different formulations. Pain/tenderness/soreness (24.8 to 28.9%) and erythema (18.4 to 21.0%) were the most commonly reported local reactions. The most common systemic adverse

events (reported by $\geq 10\%$ of subjects in one or more treatment groups, irrespective of causal relationship to vaccination) were: fever $\geq 102.0^{\circ}\text{F}$ (38.9°C) oral equivalent (27.0 to 29.2%), upper respiratory infection (26.9 to 29.7%), otitis media (12.0 to 14.1%), cough (11.0 to 15.1%), rhinorrhea (8.7 to 10.6%), and irritability (6.5 to 11.9%). Six subjects reported serious adverse events.

Two-Dose Regimen in Children

Nine hundred eighty one (981) subjects in a clinical trial received 2 doses of varicella vaccine (Oka/Merck) 3 months apart and were actively followed for 42 days after each dose. The 2-dose regimen of varicella vaccine had a safety profile comparable to that of the 1-dose regimen. The overall incidence of injection-site clinical complaints (primarily erythema and swelling) observed in the first 4 days following vaccination was 25.4% Postdose 2 and 21.7% Postdose 1, whereas the overall incidence of systemic clinical complaints in the 42-day follow-up period was lower Postdose 2 (66.3%) than Postdose 1 (85.8%).

Adolescents and Adults 13 Years of Age and Older

In clinical trials involving approximately 1600 healthy adolescents and adults, the majority of whom received two doses of varicella vaccine (Oka/Merck) and were monitored for up to 42 days after any dose, fever, injection-site complaints, or rashes were reported in decreasing order of frequency as follows: injection-site complaints (soreness, erythema, swelling, rash, pruritus, pyrexia, hematoma, induration, numbness); fever $\geq 100^{\circ}\text{F}$ (37.8°C) oral; varicella-like rash (generalized, median 5 lesions); varicella-like rash (injection site, median 2 lesions).

In a randomized open-label clinical trial (NCT00432523), conducted in France and Germany, 752 children 12 months through 18 months of age received M-M-R II concomitantly administered with VARIVAX at a separate site, by either the intramuscular (n=374) or subcutaneous (n=378) route. In the overall population, 55.3% were male and the median age was 13.2 months. Local and systemic solicited adverse reactions were recorded by parents or guardians using standardized diary cards. Local solicited reactions were recorded for 4 days after vaccination, and systemic solicited adverse reactions were recorded for 42 days after vaccination. In the event that a participant experienced a rash or a mumps-like illness, parents and/or guardians were instructed to contact the investigator for an examination as soon as possible and no later than 72 hours following onset of symptoms. The nature of any rash was characterized by principal investigator

either as measles-like, rubella-like, varicella-like or “other”. Study investigators reviewed the diary card with the participant or participant’s legal guardian 42 days after vaccination to ensure consistency with protocol definitions. Table 3 below presents the frequency of solicited adverse reactions based on the final assessment by the study investigators.

Table 3: Proportion of Participants Reporting Solicited Adverse Reactions Following Vaccination with VARIVAX Concomitantly Administered with M-M-R II, by the Intramuscular or Subcutaneous Route

	Intramuscular N=374 %	Subcutaneous N=376 %
Solicited local reactions at Varivax injection site (Days 0 to 4)*		
Erythema†	8.8	16.8
Mild	8.0	12.8
Moderate	0.5	3.7
Severe	0	0
Missing	0.3	0.3
Pain‡	7.0	8.5
Mild	4.8	7.2
Moderate	2.1	1.3
Severe	0	0
Swelling†	3.2	4.8
Mild	1.6	3.5
Moderate	1.1	0.5
Severe	0	0
Missing	0.5	0.8
Solicited systemic reactions (Days 0 to 42)		
Measles-like rash (Days 0 to 42)§	2.9	2.7
Rubella-like rash (Days 0 to 42)§	2.7	2.7
Varicella-like rash (Days 0 to 42)§	0.5	3.2
Mumps-like illness (Days 0 to 42)	0	0.3
Fever (temperature ≥38.0°C) (Days 0 to 42)¶,¶	66.5	66.8

38.0-38.5°C	20.4	22.2
>38.5-39.0°C	17.4	16.6
>39.0-39.5°C	14.2	13.4
>39.5-40.0°C	11.8	11.0
>40.0°C	2.7	3.7

N=total number of participants in the group

* During the post vaccination monitoring period (0-42 days), eight participants experienced a varicella-like injection-site rash at the VARIVAX injection site. All were reported in the subcutaneous group.

† Intensity of injection site reaction: mild or ≤ 2.5 cm; moderate or >2.5 to ≤ 5.0 cm; severe or >5.0 cm.

‡ Intensity of pain: mild: awareness of symptom but easily tolerated; moderate: definitely acting like something is wrong; severe: extremely distressed or unable to do usual activities.

§ Testing to distinguish between rash caused by wild-type or vaccine virus was not performed. Reports of measles-, rubella-, and varicella-like rash included 3 reports of measles, 1 report of rubella, and 1 report of varicella, all with onset within 15 days post-vaccination.

¶ The percentage of fever is defined within the population who had valid temperature measurements. One participant in IM group and two participants in SC group did not have temperature measurements and were excluded from the denominator; resulting in N=373 and N=374, respectively.

In the IM Group 92.3% of fevers were documented using the rectal route of measurement and 7.7% of fevers were documented only by the axillary route of measurement. In the SC Group 89.6% of fevers were documented using the rectal route of measurement and 10.4% of fevers were documented only by the axillary route of measurement.

Unsolicited adverse events that occurred within 42 days following vaccination were recorded using diary cards supplemented by medical review. Data on unsolicited adverse events were transcribed into the study database during an on-site visit at day 42. The rates and types of reported adverse events (AEs) across groups were similar and included common clinical events that are often reported in the evaluated populations. Serious adverse events occurred at rates of 0.3% and 1% in the intramuscular and subcutaneous groups, respectively. One moderate intensity case of otitis media occurred in a participant in the subcutaneous group was considered related to the vaccination.

The following additional side effects have been reported regardless of causality since the vaccine has been marketed:

Body As A Whole: Anaphylaxis (including anaphylactic shock) and related phenomena such as angioneurotic edema, facial edema, and peripheral edema; anaphylaxis in individuals with or without an allergic history.

Eye Disorders: Necrotizing retinitis (reported only in immunocompromised individuals).

Gastrointestinal Disorders: Nausea; vomiting.

Hemic and Lymphatic System: Aplastic anemia; thrombocytopenia (including idiopathic thrombocytopenic purpura (ITP)), lymphadenopathy.

Infections and Infestations: Varicella (vaccine strain).

Nervous/Psychiatric: Encephalitis[†]; cerebrovascular accident; transverse myelitis; Guillain-Barré syndrome; Bell's palsy; ataxia; febrile and non-febrile seizures; aseptic meningitis; meningitis[†]; dizziness; paresthesia; irritability; syncope.

Respiratory: Pharyngitis; pneumonia/pneumonitis; upper respiratory tract infection.

Skin: Stevens-Johnson syndrome; erythema multiforme; Henoch-Schönlein purpura; secondary bacterial infections of skin and soft tissue, including cellulitis; herpes zoster[†].

[†]Cases caused by wild-type varicella or vaccine strain varicella have been reported in immunocompromised and immunocompetent individuals.

XV. OVERDOSAGE

There are no data with regard to overdose.

XVI. APPEARANCE AND AVAILABILITY

Lyophilized white, compact, crystalline pellet.

VARIVAX Refrigerated is supplied as a single dose vial of lyophilized vaccine and a vial of diluent.

XVII. STORAGE

Stability

VARIVAX Refrigerated has a minimum potency level of approximately 1350 PFU 30 minutes after reconstitution at room temperature (20 to 25°C, 68 to 77°F).

Storage

Vaccine Vial

During shipment, to ensure that there is no loss of potency, the vaccine must be maintained at a temperature of 2 to 8°C or colder (36 to 46°F or colder), but not exceed temperatures lower than -50°C (-58°F). Use of dry ice may subject VARIVAX to temperatures colder than -50°C (-58°F).

Before reconstitution, VARIVAX Refrigerated has a shelf-life of 24 months when refrigerated at 2 to 8°C or colder (36 to 46°F or colder). The vaccine may also be stored in a freezer; if subsequently transferred to a refrigerator, **THE VACCINE SHOULD NOT BE REFROZEN.**

Before reconstitution, protect from light.

DISCARD IF RECONSTITUTED VACCINE IS NOT USED WITHIN 30 MINUTES.

Combination pack with vaccine vial and diluent:

For combination packs with vaccine vial and diluent packaged together, store in the refrigerator at 2 to 8°C (36 to 46°F). **DO NOT STORE THE COMBINATION PACK IN THE FREEZER.**

XVIII. SHELF LIFE

Please refer to expiry date on the outer carton.

XIX. MANUFACTURER

MERCK SHARP & DOHME LLC

Old Oxford Road Durham,
NC 27712, U.S.A.

XX. PACKER AND RELEASER

MERCK SHARP & DOHME B.V.

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XXI. PRODUCT REGISTRATION HOLDER

MERCK SHARP & DOHME (MALAYSIA) SDN. BHD.

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